

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Topic

Topics: Linear Systems & LU, PCA, Locking Protocols (2PL), Logical Reasoning (Series & Coding)

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1 Topic 1: Linear Equations, Gaussian Elimination & LU Decomposition

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | NAT | 2 Marks]

Consider the following system of linear equations:

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\x + 2y + \lambda z &= \mu\end{aligned}$$

For what value of λ does the system have either no solution or infinitely many solutions? **Answer:**

3 Solution: Write the augmented matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix}$$

Subtract row 1 from rows 2 and 3:

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{bmatrix}$$

Subtract row 2 from row 3:

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{bmatrix}$$

If $\lambda - 3 = 0 \implies \lambda = 3$, the rank of coefficient matrix is 2. If $\mu = 10$, rank of augmented matrix is 2 (infinitely many solutions). If $\mu \neq 10$, rank is 3 (no solution). Thus, $\lambda = 3$.

Q1.2

[GATE DA 2024 | MCQ | 1 Mark]

Let A be a non-singular 3×3 matrix with an LU decomposition where L is unit lower triangular and U is upper triangular. If the diagonal entries of U are 2, -3 , and 5, what is $\det(A)$?

- (A) -30
- (B) 30
- (C) 0
- (D) 1

Answer: (A) Solution: $A = LU$. Therefore, $\det(A) = \det(L) \det(U)$. Since L is unit lower triangular, its diagonal entries are all 1, so $\det(L) = 1$. The determinant of upper triangular matrix U is the product of its diagonal entries: $\det(U) = 2 \times (-3) \times 5 = -30$. Hence $\det(A) = 1 \times (-30) = -30$.

Q1.3

[GATE CS 2023 | NAT | 2 Marks]

For the matrix $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$, let its LU decomposition be $A = LU$ where $L = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix}$ and $U = \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$. What is the value of u_{22} ? **Answer: 3 Solution:** Multiply L and U :

$$\begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$$

Comparing entries: $u_{11} = 2, u_{12} = 1$. Then $l_{21}u_{11} = 4 \implies l_{21}(2) = 4 \implies l_{21} = 2$. Next, $l_{21}u_{12} + u_{22} = 5 \implies (2)(1) + u_{22} = 5 \implies u_{22} = 3$.

Q1.4

[GATE DA 2025 | MSQ | 2 Marks]

Consider a linear system $Ax = b$ where A is an $m \times n$ real matrix. Which of the following statements is/are TRUE?

- (A) If $\text{rank}(A) = \text{rank}([A \mid b])$, the system is consistent.
- (B) If $\text{rank}(A) = \text{rank}([A \mid b]) = n$, the system has a unique solution.
- (C) If $m < n$ and the system is consistent, it must have infinitely many solutions.
- (D) If $m > n$, the system cannot be consistent.

Answer: (A, B, C) **Solution:** By the Rouché-Capelli Theorem: (A) Consistency holds if and only if $\text{rank}(A) = \text{rank}([A \mid b])$. (B) If this rank equals the number of variables n , there are no free variables, yielding a unique solution. (C) If $m < n$, nullity is at least $n - m > 0$, giving infinitely many solutions for any consistent system. (D) is false: an overdetermined system can be consistent if redundant equations are linear combinations of others.

Q1.5

[GATE CS 2021 | NAT | 2 Marks]

How many floating-point multiplication and division operations (flops) are required in the forward elimination phase of Gaussian elimination for an $n \times n$ system without pivoting, as a function of the leading term? If $n = 10$, what is $\frac{n^3}{3}$ to the nearest integer? **Answer: 333** **Solution:** Forward elimination requires approximately $\sum_{k=1}^{n-1} (n-k)(n-k+1) \approx \frac{n^3}{3}$ operations. For $n = 10$, $\frac{10^3}{3} = \frac{1000}{3} \approx 333$.

Q1.6

[GATE CS 2020 | MCQ | 1 Mark]

Gaussian elimination with partial pivoting interchanges rows to ensure that:

- (A) The pivot element has the largest absolute value in the current column on or below the diagonal.
- (B) The determinant of the matrix becomes 1.
- (C) All diagonal elements become positive.
- (D) No zero entries appear in the lower triangular factor L .

Answer: (A) **Solution:** Partial pivoting swaps rows so that the pivot element chosen in step k satisfies $|a_{ik}| = \max_{j \geq k} |a_{jk}|$, minimizing numerical division errors and roundoff error propagation.

Q1.7

[GATE DA 2026 | NAT | 1 Mark]

Consider the homogeneous system of equations:

$$\begin{aligned} x + 2y - z &= 0 \\ 2x + 4y + kz &= 0 \end{aligned}$$

For what value of k does the dimension of the solution space equal 2? **Answer: -2** **Solution:** Subtract $2 \times$ equation 1 from equation 2: $(2-2)x + (4-4)y + (k-(-2))z = 0 \implies (k+2)z = 0$. If $k = -2$, the second equation is completely redundant ($0 = 0$), leaving 1 independent equation in 3 variables. By the Rank-Nullity Theorem, the dimension of the solution space is $n - r = 3 - 1 = 2$. Thus $k = -2$.

Q1.8

[GATE CS 2019 | MCQ | 2 Marks]

An LU decomposition of a square matrix A without row interchanges exists if and only if:

- (A) All leading principal submatrices of A are non-singular.
- (B) A is symmetric and positive definite.
- (C) All eigenvalues of A are strictly positive.
- (D) $\det(A) > 0$.

Answer: (A) **Solution:** A matrix A has a unique Doolittle/Crout LU factorization without pivoting if and only if all of its leading principal minors (the determinants of its $k \times k$ top-left submatrices for $k = 1, \dots, n - 1$) are non-zero.

Q1.9

[GATE DA 2025 | NAT | 2 Marks]

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. If $PA = LU$ where P is a permutation matrix swapping row 1 and row 2, and L is unit lower triangular, what is the entry u_{11} of U ? **Answer: 3** **Solution:** Swapping rows:

$$PA = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

In LU decomposition of PA : $L = \begin{bmatrix} 1 & 0 \\ 1/3 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 3 & 4 \\ 0 & 2 - (1/3)(4) \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 0 & 2/3 \end{bmatrix}$. The entry $u_{11} = 3$.

Q1.10

[GATE CS 2018 | MCQ | 1 Mark]

If A is a symmetric positive definite matrix, its factorization $A = LL^T$ where L is lower triangular with positive diagonal elements is called:

- (A) Cholesky decomposition
- (B) QR decomposition
- (C) Singular Value Decomposition
- (D) Spectral decomposition

Answer: (A) **Solution:** By definition, factoring a symmetric positive definite matrix into LL^T (where L is lower triangular with positive diagonal entries) is the Cholesky factorization, requiring half the computation of general LU .

1.2 10 Practice Questions (GATE Level)

P1.1

[Practice | NAT | 2 Marks]

For what value of c does the system below have NO solution?

$$2x + 3y = 8$$

$$4x + 6y = c$$

If any $c \neq 16$ gives no solution, what is the value of c for which infinitely many solutions exist?

Answer: 16 **Solution:** Multiply the first equation by 2: $4x + 6y = 16$. For the system to be consistent with redundant equations (infinitely many solutions), the right-hand sides must match: $c = 16$. For any $c \neq 16$, the lines are parallel and distinct, resulting in no solution.

P1.2

[Practice | MCQ | 1 Mark]

In solving $Ax = b$ using LU decomposition ($A = LU$), the two triangular systems solved sequentially are:

- (A) $Ly = b$ (forward substitution) followed by $Ux = y$ (back substitution)

- (B) $Uy = b$ followed by $Lx = y$
 (C) $Lx = b$ followed by $Ux = b$
 (D) $LUx = b$ via direct matrix inversion

Answer: (A) **Solution:** $Ax = b \implies L(Ux) = b$. Setting $y = Ux$, we first solve $Ly = b$ using forward substitution, and then solve $Ux = y$ using backward substitution.

P1.3

[Practice | NAT | 2 Marks]

Let $A = \begin{bmatrix} 4 & 2 \\ 2 & 10 \end{bmatrix}$. If $A = LL^T$ is the Cholesky decomposition with $L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$, what is the value of l_{22} ? **Answer: 3** **Solution:** $LL^T = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 10 \end{bmatrix}$. From (1, 1): $l_{11}^2 = 4 \implies l_{11} = 2$. From (2, 1): $l_{11}l_{21} = 2 \implies 2l_{21} = 2 \implies l_{21} = 1$. From (2, 2): $l_{21}^2 + l_{22}^2 = 10 \implies 1^2 + l_{22}^2 = 10 \implies l_{22}^2 = 9 \implies l_{22} = 3$.

P1.4

[Practice | MSQ | 2 Marks]

Which of the following conditions implies that the linear system $Ax = b$ has a unique solution?

- (A) A is square and $\det(A) \neq 0$.
 (B) A is $m \times n$ with $m = n = \text{rank}(A)$.
 (C) The null space of A contains only the zero vector and $b \in \text{Col}(A)$.
 (D) A has more columns than rows ($n > m$).

Answer: (A, B, C) **Solution:** (A) Non-singular square matrix ensures A^{-1} exists, so $x = A^{-1}b$ is unique. (B) Full rank square matrix. (C) Trivial null space means uniqueness, and $b \in \text{Col}(A)$ ensures existence. (D) Underdetermined systems ($n > m$) have at least one free variable, so they cannot have a unique solution.

P1.5

[Practice | NAT | 1 Mark]

What is the determinant of the lower triangular matrix $L = \begin{bmatrix} 3 & 0 & 0 \\ -1 & 4 & 0 \\ 5 & 2 & 2 \end{bmatrix}$? **Answer: 24** **Solution:** The determinant of a triangular matrix is the product of its diagonal elements: $\det(L) = 3 \times 4 \times 2 = 24$.

P1.6

[Practice | MCQ | 2 Marks]

Consider applying Gaussian elimination without pivoting to the matrix $\begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$. The algorithm fails at the first step because:

- (A) The pivot element a_{11} is zero, causing division by zero.
 (B) The determinant is negative.
 (C) The matrix is non-invertible.
 (D) The rows are linearly dependent.

Answer: (A) **Solution:** The multiplier used to eliminate entries below the pivot is $m_{21} = \frac{a_{21}}{a_{11}}$. Because $a_{11} = 0$, the standard algorithm without pivoting encounters a division by zero error, requiring a row swap (pivoting).

P1.7

[Practice | NAT | 2 Marks]

In the system $Ax = b$ where $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ and $b = \begin{bmatrix} 5 \\ 12 \end{bmatrix}$, find the solution value for x_1 . **Answer:**

-1 Solution: Subtract $2 \times$ row 1 from row 2: $(7 - 6)x_2 = 12 - 10 \implies x_2 = 2$. Back-substituting into row 1: $x_1 + 3(2) = 5 \implies x_1 + 6 = 5 \implies x_1 = -1$.

P1.8

[Practice | MSQ | 2 Marks]

Which of the following operations are elementary row operations that preserve the solution set of a system of linear equations?

- (A) Interchanging two rows.
- (B) Multiplying a row by a non-zero scalar.
- (C) Adding a scalar multiple of one row to another row.
- (D) Squaring all entries in a row.

Answer: (A, B, C) Solution: (A), (B), and (C) are the three standard elementary row operations (which correspond to multiplication by invertible elementary matrices). (D) is a non-linear operation that alters the solution set and introduces extraneous roots.

P1.9

[Practice | NAT | 1 Mark]

A system $Ax = 0$ with 4 equations and 6 unknowns has $\text{rank}(A) = 3$. What is the number of free variables in the solution? **Answer: 3 Solution:** Number of variables $n = 6$. Rank $r = 3$. The number of free variables equals the nullity: $\text{nullity} = n - r = 6 - 3 = 3$.

P1.10

[Practice | MCQ | 1 Mark]

If an upper triangular matrix U has a zero on its main diagonal, then:

- (A) $\det(U) = 0$ and U is singular.
- (B) U must be the zero matrix.
- (C) U has only zero eigenvalues.
- (D) $U^T U$ is strictly positive definite.

Answer: (A) Solution: Because $\det(U) = \prod_{i=1}^n u_{ii}$, if any diagonal element is zero, the product is zero, meaning $\det(U) = 0$ and U is not invertible.

2 Topic 2: PCA – Dimensionality Reduction (Math + Application)

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | NAT | 2 Marks]

Let the sample covariance matrix of a 2D dataset be $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$. What is the percentage of total variance explained by the first principal component? **Answer: 75** **Solution:** Eigenvalues of Σ : $\det(\Sigma - \lambda I) = (4 - \lambda)^2 - 4 = \lambda^2 - 8\lambda + 12 = (\lambda - 6)(\lambda - 2) = 0 \implies \lambda_1 = 6, \lambda_2 = 2$. Total variance is $\text{trace}(\Sigma) = \lambda_1 + \lambda_2 = 6 + 2 = 8$. The variance explained by the first principal component is $\frac{\lambda_1}{\lambda_1 + \lambda_2} \times 100\% = \frac{6}{8} \times 100\% = 75\%$.

Q2.2

[GATE DA 2024 | MCQ | 1 Mark]

In Principal Component Analysis (PCA), the principal component directions correspond to:

- (A) Eigenvectors of the sample covariance matrix.
- (B) Eigenvalues of the feature correlation matrix only.
- (C) Left singular vectors of the centered data matrix divided by n .
- (D) The centroids of k -means clusters.

Answer: (A) **Solution:** PCA finds orthonormal projection directions that maximize variance, which are mathematically identical to the eigenvectors of the data sample covariance matrix $\Sigma = \frac{1}{n-1} X^T X$ (where X is zero-centered).

Q2.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements about Principal Component Analysis (PCA) is/are TRUE?

- (A) The principal components are mutually orthogonal.
- (B) The first principal component points in the direction of maximum sample variance.
- (C) PCA is sensitive to the scale of the original features.
- (D) PCA uses class label information to maximize separation between different classes.

Answer: (A, B, C) **Solution:** (A) The covariance matrix is symmetric, so its eigenvectors can be chosen to be mutually orthogonal. (B) By definition, PC1 maximizes variance. (C) Features with larger numeric scales dominate covariance; hence standardization is necessary. (D) is false because PCA is completely unsupervised (Linear Discriminant Analysis uses labels).

Q2.4

[GATE CS 2021 | NAT | 2 Marks]

A dataset with 3 features has covariance matrix with eigenvalues $\lambda_1 = 12$, $\lambda_2 = 5$, and $\lambda_3 = 3$. What is the proportion of total variance explained by the top two principal components combined?

Answer: 0.85 **Solution:** Total variance = $\lambda_1 + \lambda_2 + \lambda_3 = 12 + 5 + 3 = 20$. Sum of top two eigenvalues = $\lambda_1 + \lambda_2 = 12 + 5 = 17$. The proportion of variance explained is $\frac{17}{20} = 0.85$.

Q2.5

[GATE DA 2025 | MCQ | 1 Mark]

Let X be an $n \times d$ centered data matrix with Singular Value Decomposition (SVD) $X = USV^T$. The principal component loading vectors (directions) are given by:

- (A) The columns of V
- (B) The columns of U
- (C) The diagonal entries of S
- (D) The rows of U

Answer: (A) **Solution:** The sample covariance matrix is $\Sigma \propto X^T X = (VSU^T)(USV^T) = VS^2V^T$. Thus, the eigenvectors of $X^T X$ are the columns of the right singular matrix V .

Q2.6

[GATE CS 2022 | NAT | 1 Mark]

Let X be a centered dataset in $\mathbb{R}^2$. If the first principal component is $v_1 = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]^T$, what is the slope of the line representing the second principal component? **Answer: -1** **Solution:** In 2D, the two principal component directions must be mutually orthogonal. If $v_1 = [1/\sqrt{2}, 1/\sqrt{2}]^T$ (slope $m_1 = 1$), the orthogonal vector is $v_2 = [-1/\sqrt{2}, 1/\sqrt{2}]^T$ (or $[1/\sqrt{2}, -1/\sqrt{2}]^T$). Its slope is $m_2 = \frac{1/\sqrt{2}}{-1/\sqrt{2}} = -1$.

Q2.7

[GATE DA 2026 | MCQ | 2 Marks]

Suppose PCA is applied to reduce the dimensionality of a dataset from d dimensions to k dimensions ($k < d$). The reconstruction error (mean squared error between original and projected points) is equal to:

- (A) The sum of the discarded eigenvalues $\sum_{j=k+1}^d \lambda_j$
- (B) The sum of the retained eigenvalues $\sum_{j=1}^k \lambda_j$
- (C) The product of all eigenvalues $\prod_{j=1}^d \lambda_j$
- (D) Zero for all values of k

Answer: (A) **Solution:** By the Eckart-Young-Mirsky Theorem, project-reconstruct error using k principal components is exactly the sum of the variances along the discarded orthogonal axes: $\sum_{j=k+1}^d \lambda_j$.

Q2.8

[GATE CS 2023 | MSQ | 2 Marks]

Let X be a zero-centered dataset with covariance matrix Σ . Let $Y = XV$ be the transformed coordinates (principal component scores). Which of the following is/are TRUE?

- (A) The sample covariance matrix of Y is a diagonal matrix.
- (B) The variance of the j -th column of Y is λ_j .
- (C) The sum of variances of the columns of Y equals the sum of variances of the columns of X .
- (D) The columns of Y have non-zero sample correlations with each other.

Answer: (A, B, C) **Solution:** (A) $\text{Cov}(Y) = V^T \Sigma V = \Lambda$ is diagonal, meaning the transformed variables are uncorrelated (D is false). (B) The diagonal entries are the eigenvalues λ_j . (C) Trace is invariant under orthogonal rotation: $\text{trace}(\Sigma) = \text{trace}(\Lambda)$.

Q2.9

[GATE DA 2024 | NAT | 1 Mark]

If a 5-dimensional dataset has total variance equal to 50, and the first 3 principal components account for 42 units of variance, what percentage of the variance is lost after dimensionality reduction to 3 dimensions? **Answer: 16** **Solution:** The discarded variance is $50 - 42 = 8$. Percentage of variance lost is $\frac{8}{50} \times 100\% = 16\%$.

Q2.10

[GATE CS 2020 | MCQ | 1 Mark]

Prior to performing PCA, why is it standard practice to center the data by subtracting the sample mean from each feature?

- (A) To ensure that the first principal component captures the direction of maximum variance rather than the displacement from the origin.
- (B) To ensure that all eigenvalues are strictly positive integers.

- (C) To make the covariance matrix asymmetric.
- (D) To eliminate the need for computing eigenvectors.

Answer: (A) **Solution:** If the data is not centered, the first component tends to point toward the sample mean vector rather than along the axis of greatest variation.

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | NAT | 2 Marks]

The covariance matrix of two features is $\Sigma = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$. What is the largest eigenvalue λ_1 ? **An-**

swer: 8 **Solution:** Characteristic equation: $(5 - \lambda)^2 - 9 = 0 \implies 5 - \lambda = \pm 3 \implies \lambda = 5 \pm 3$. Thus $\lambda_1 = 8$ and $\lambda_2 = 2$. The largest eigenvalue is 8.

P2.2

[Practice | MCQ | 1 Mark]

If all features in a dataset are completely uncorrelated and have identical variance σ^2 , what are the principal components?

- (A) Any set of orthonormal vectors forms a valid set of principal components.
- (B) There is only one unique principal component.
- (C) PCA cannot be applied because the covariance matrix is undefined.
- (D) The first principal component explains 100% of the variance.

Answer: (A) **Solution:** If $\Sigma = \sigma^2 I$, all eigenvalues are equal to σ^2 . The eigenspace is the entire space $\mathbb{R}^d$, meaning any orthogonal basis is an eigenbasis and no single direction has greater variance than any other.

P2.3

[Practice | NAT | 2 Marks]

A dataset with $n = 100$ samples and $d = 20$ features is transformed using PCA. If the eigenvalues of the sample covariance matrix are $\lambda_1 = 10, \lambda_2 = 6, \lambda_3 = 4$, and $\lambda_4 = \dots = \lambda_{20} = 0$, what is the intrinsic dimensionality (rank) of the data? **Answer: 3** **Solution:** The number of non-zero eigenvalues equals the rank of the covariance matrix. Here, exactly 3 eigenvalues are positive ($\lambda_1, \lambda_2, \lambda_3$), so the data lies strictly on a 3-dimensional subspace.

P2.4

[Practice | MSQ | 2 Marks]

Which of the following statements about Kernel PCA is/are TRUE?

- (A) It allows finding non-linear principal component manifolds.
- (B) It applies the kernel trick to perform PCA in a high-dimensional feature space $\Phi(x)$.
- (C) It requires computing the explicit coordinates of each point in feature space.
- (D) The kernel matrix K must be centered before computing its eigenvectors.

Answer: (A, B, D) **Solution:** (A) Kernel PCA captures non-linear structures. (B) It uses $K(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$. (C) is false: the kernel trick avoids computing $\Phi(x)$ explicitly. (D) Centering in feature space is carried out implicitly on the kernel Gram matrix K .

P2.5

[Practice | NAT | 2 Marks]

Let a centered data point be $x = [2, 1]^T$. If the unit-length first principal component is $v_1 = [0.8, 0.6]^T$, what is the projected coordinate (score) of x along the first principal component? **Answer: 2.20** **Solution:** The projection of x onto v_1 is the scalar dot product: $z_1 = x^T v_1 = (2)(0.8) + (1)(0.6) = 1.6 + 0.6 = 2.20$.

P2.6**[Practice | MCQ | 1 Mark]**

A scree plot in PCA is a visual diagnostic plot showing:

- (A) Eigenvalues (or variance explained) ordered from largest to smallest versus component index.
- (B) Residual error versus learning rate.
- (C) False positive rate versus true positive rate.
- (D) Cluster centroids versus number of iterations.

Answer: (A) Solution: A scree plot displays the eigenvalues associated with each component in descending order to identify the “elbow” point for deciding how many components to retain.

P2.7**[Practice | NAT | 1 Mark]**

If the eigenvalues of Σ are $\lambda_1 = 40, \lambda_2 = 30, \lambda_3 = 20, \lambda_4 = 10$, what is the minimum number of principal components needed to explain at least 85% of the total variance? **Answer: 3 Solution:** Total variance = $40 + 30 + 20 + 10 = 100$. With 1 component: 40%. With 2 components: $40 + 30 = 70\%$. With 3 components: $40 + 30 + 20 = 90\% \geq 85\%$. Thus, 3 components are needed.

P2.8**[Practice | MCQ | 2 Marks]**

How does PCA differ from Linear Discriminant Analysis (LDA)?

- (A) PCA is unsupervised and maximizes overall variance; LDA is supervised and maximizes class separability.
- (B) PCA requires class labels; LDA does not.
- (C) PCA is non-linear; LDA is strictly linear.
- (D) PCA only works on 2D data.

Answer: (A) Solution: PCA does not use class labels and finds axes of maximum variance, whereas LDA is a supervised technique that maximizes between-class variance relative to within-class variance.

P2.9**[Practice | NAT | 2 Marks]**

A data matrix X of size 50×10 has rank 6. What is the maximum number of non-zero eigenvalues that its sample covariance matrix $\Sigma = \frac{1}{49} X^T X$ can have? **Answer: 6 Solution:** $\text{rank}(X^T X) = \text{rank}(X) = 6$. The number of non-zero eigenvalues of a matrix equals its rank. Therefore, there are at most 6 non-zero eigenvalues.

P2.10**[Practice | MSQ | 2 Marks]**

Which of the following operations will change the direction of the principal components?

- (A) Multiplying one feature by a constant factor of 10 while leaving other features unchanged.
- (B) Translating all data points by adding a constant vector c .
- (C) Applying an arbitrary non-uniform scaling to different features.
- (D) Rotating the entire dataset rigidly around the origin.

Answer: (A, C, D) Solution: (A) and (C) Non-uniform scaling alters feature variances, changing the principal axes. (B) Subtracting the mean removes translation, leaving eigenvectors unchanged. (D) Rigid rotation rotates the principal axes by the exact same rotation.

3 Topic 3: Concurrency Control – Locking Protocols (2PL)

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2024 | MCQ | 1 Mark]

The Two-Phase Locking (2PL) protocol guarantees which of the following properties for all generated schedules?

- (A) Conflict serializability
- (B) Freedom from deadlocks
- (C) Freedom from cascading rollbacks
- (D) View serializability but not conflict serializability

Answer: (A) **Solution:** Basic 2PL ensures that once a transaction releases any lock, it cannot acquire any new lock. This lock-point monotonicity ensures that the precedence graph is acyclic, guaranteeing conflict serializability. However, basic 2PL does not prevent deadlocks or cascading aborts.

Q3.2

[GATE CS 2023 | NAT | 2 Marks]

Consider a transaction T executing the following sequence of lock and unlock requests:

Lock-S(A); Lock-X(B); Unlock(A); Lock-S(C); Unlock(B); Unlock(C);

Does transaction T satisfy the Two-Phase Locking protocol? Enter 1 for Yes, 0 for No. **Answer: 0**

Solution: Under 2PL, all lock operations must precede all unlock operations. Here, transaction T executes Unlock(A) and subsequently acquires a new lock Lock-S(C). Since an acquisition occurs after a release, T violates the two-phase rule (0).

Q3.3

[GATE DA 2024 | MCQ | 1 Mark]

In Strict Two-Phase Locking (Strict 2PL), a transaction must hold:

- (A) All exclusive (X) locks until the transaction commits or aborts.
- (B) All shared (S) locks until the transaction commits.
- (C) Exactly two locks at any time.
- (D) Locks only during the first phase.

Answer: (A) **Solution:** Strict 2PL requires that all exclusive (write) locks taken by a transaction be held until the transaction finishes (commit or abort), ensuring schedules are cascadeless and recoverable.

Q3.4

[GATE CS 2022 | MSQ | 2 Marks]

Which of the following statements comparing 2PL variants is/are TRUE?

- (A) Rigorous 2PL requires both shared and exclusive locks to be held until transaction commit/abort.
- (B) Strict 2PL schedules avoid cascading aborts.
- (C) Rigorous 2PL produces schedules whose serialization order is the same as transaction commit order.
- (D) Basic 2PL guarantees freedom from deadlocks.

Answer: (A, B, C) **Solution:** (A) Rigorous 2PL holds all locks (S and X) until the end of transaction. (B) Strict 2PL prevents dirty reads, ensuring cascadelessness. (C) Holding all locks until commit makes commit order equal serialization order. (D) is false: 2PL allows wait-for cycles (deadlocks).

Q3.5

[GATE DA 2025 | MCQ | 1 Mark]

A lock conversion in 2PL that changes a shared lock to an exclusive lock is called:

- (A) Lock upgrading
- (B) Lock downgrading
- (C) Lock escalation
- (D) Lock de-escalation

Answer: (A) **Solution:** Upgrading converts a shared (read) lock to an exclusive (write) lock, and can only occur during the growing phase. Downgrading converts an exclusive lock to a shared lock during the shrinking phase.

Q3.6

[GATE CS 2021 | NAT | 2 Marks]

In a database with two transactions, T_1 acquires Lock-X(A) and requests Lock-X(B). Simultaneously, T_2 acquires Lock-X(B) and requests Lock-X(A). Both transactions follow 2PL. How many transactions are involved in the resulting deadlock cycle? **Answer: 2** **Solution:** T_1 holds A and waits for T_2 to release B ($T_1 \rightarrow T_2$). T_2 holds B and waits for T_1 to release A ($T_2 \rightarrow T_1$). The Wait-For Graph contains a cycle $T_1 \Rightarrow T_2$, involving exactly 2 transactions.

Q3.7

[GATE CS 2020 | MCQ | 2 Marks]

Conservative 2PL (Static 2PL) prevents deadlocks by requiring that a transaction:

- (A) Pre-declares and acquires all needed locks simultaneously before beginning execution.
- (B) Releases all locks whenever another transaction requests them.
- (C) Uses only shared locks.
- (D) Never writes to disk.

Answer: (A) **Solution:** Conservative 2PL eliminates the “hold and wait” deadlock condition by acquiring all required locks upfront before the transaction starts execution. If any lock is unavailable, it acquires none and waits.

Q3.8

[GATE DA 2026 | MSQ | 2 Marks]

Which of the following schedules can be generated by a scheduler following the Strict 2PL protocol?

- (A) Conflict serializable schedules
- (B) Recoverable schedules
- (C) Cascadeless schedules
- (D) Schedules with cascading rollbacks

Answer: (A, B, C) **Solution:** Strict 2PL is a subset of 2PL, so it guarantees conflict serializability (A). Because exclusive locks are held until commit/abort, other transactions cannot read uncommitted writes, ensuring recoverability (B) and eliminating cascading rollbacks (C). Thus (D) is false.

Q3.9

[GATE CS 2019 | MCQ | 1 Mark]

In multiple granularity locking, which intention lock must be acquired on the parent node before acquiring an exclusive lock on a child node?

- (A) Intention Exclusive (IX)
- (B) Intention Shared (IS)
- (C) Shared (S)
- (D) Shared Intention Exclusive (SIX)

Answer: (A) **Solution:** To set an exclusive lock on a lower-level node (e.g., tuple), the transaction must traverse down the hierarchy and set Intention Exclusive (IX) locks on all ancestor nodes (tables, databases).

Q3.10

[GATE CS 2018 | NAT | 1 Mark]

If the compatibility matrix of shared (S) and exclusive (X) locks has 4 entries $((S, S), (S, X), (X, S), (X, X))$, how many of these combinations are compatible (allow concurrent acquisition)? **Answer: 1** **Solution:** Only the shared-shared combination (S, S) is compatible (TRUE). (S, X) , (X, S) , and (X, X) are incompatible (FALSE). Thus, exactly 1 combination is compatible.

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | MCQ | 1 Mark]

What distinguishes Rigorous 2PL from Strict 2PL?

- (A) Rigorous 2PL holds both shared and exclusive locks until end-of-transaction, whereas Strict 2PL holds only exclusive locks until end-of-transaction.
- (B) Rigorous 2PL does not allow shared locks.
- (C) Strict 2PL allows deadlocks, but Rigorous 2PL prevents them.
- (D) Rigorous 2PL releases locks during the growing phase.

Answer: (A) **Solution:** Strict 2PL allows shared locks to be released before commit (during the shrinking phase), holding only exclusive locks until commit. Rigorous 2PL requires all locks (both shared and exclusive) to be held until commit/abort.

P3.2

[Practice | NAT | 2 Marks]

Consider transactions T_1 and T_2 following 2PL. T_1 acquires its final lock at time $t = 5$, and T_2 acquires its final lock at time $t = 8$. In the conflict equivalent serial schedule, which transaction must appear first? Enter 1 for T_1 , 2 for T_2 . **Answer: 1** **Solution:** In 2PL, the point where a transaction acquires its last lock is called its lock point. Transactions serialize in the order of their lock points. Since T_1 's lock point ($t = 5$) precedes T_2 's lock point ($t = 8$), T_1 serializes before T_2 (1).

P3.3

[Practice | MSQ | 2 Marks]

Which of the following locking protocols guarantee conflict serializability?

- (A) Basic Two-Phase Locking
- (B) Strict Two-Phase Locking
- (C) Rigorous Two-Phase Locking
- (D) Graph-based locking (Tree protocol)

Answer: (A, B, C, D) **Solution:** Basic 2PL, Strict 2PL, and Rigorous 2PL all enforce two phases and guarantee conflict serializability. The tree-based protocol on directed acyclic database graphs also guarantees conflict serializability and deadlock freedom.

P3.4

[Practice | NAT | 1 Mark]

In a Wait-For Graph (WFG) used for deadlock detection, vertices represent: enter 1 for Data Items, or 2 for Active Transactions. **Answer: 2** **Solution:** In a Wait-For Graph, each vertex corresponds to an active transaction. A directed edge $T_i \rightarrow T_j$ indicates that T_i is waiting for a lock currently held by T_j . Thus, vertices represent transactions (2).

P3.5

[Practice | MCQ | 1 Mark]

In the Wait-Die deadlock prevention scheme based on timestamps (where smaller timestamp means older transaction), if transaction T_i requests a lock held by T_j :

- (A) If T_i is older than T_j ($TS(T_i) < TS(T_j)$), T_i is allowed to wait; otherwise T_i dies (aborts).
- (B) If T_i is older, it wounds (aborts) T_j .
- (C) T_i always waits regardless of timestamp.
- (D) Both transactions are immediately aborted.

Answer: (A) **Solution:** In Wait-Die: older transactions are allowed to wait for younger ones ($TS(T_i) < TS(T_j) \implies \text{wait}$); if a younger transaction requests a lock held by an older one ($TS(T_i) > TS(T_j)$), the younger transaction dies (aborts).

P3.6

[Practice | NAT | 2 Marks]

Consider 3 transactions T_1, T_2, T_3 running under 2PL. If the operations conflict such that $T_1 \rightarrow T_2$ and $T_2 \rightarrow T_3$, how many conflict serial orders are possible? **Answer: 1** **Solution:** The precedence relations specify T_1 before T_2 and T_2 before T_3 . The only topological ordering is $T_1 \rightarrow T_2 \rightarrow T_3$, giving exactly 1 equivalent serial order.

P3.7

[Practice | MCQ | 2 Marks]

Which of the following is an advantage of the Tree Protocol over Two-Phase Locking?

- (A) It is deadlock-free and locks can be released earlier than in 2PL.
- (B) It allows arbitrary unlocking and relocking of items.
- (C) It does not require knowledge of database item hierarchy.
- (D) It guarantees cascadelessness under all circumstances.

Answer: (A) **Solution:** The tree protocol guarantees conflict serializability and freedom from deadlocks without requiring two-phase locking, allowing early unlock of parent nodes. However, it requires a tree-structured database graph and may lock unneeded items.

P3.8

[Practice | MSQ | 2 Marks]

Which of the following conditions can cause starvation of a transaction under concurrency control?

- (A) A transaction repeatedly waiting for an exclusive lock while a stream of other transactions acquire shared locks on the same item.
- (B) The transaction being repeatedly chosen as the deadlock victim and aborted.
- (C) Using a First-Come, First-Served (FCFS) queue for lock requests.
- (D) Using Conservative 2PL when all needed locks are free.

Answer: (A, B) **Solution:** (A) Concurrency starvation occurs when readers continually jump ahead of a waiting writer. (B) Repeatedly choosing the same victim starves it. (C) FCFS lock granting guarantees fair progress and prevents starvation. (D) Free locks are granted immediately without starvation.

P3.9

[Practice | NAT | 1 Mark]

A transaction holds an exclusive lock on record R . Can another transaction obtain a shared lock on record R simultaneously? Enter 1 for Yes, 0 for No. **Answer: 0** **Solution:** Exclusive locks are incompatible with shared locks. No other transaction can acquire any lock (S or X) on R until the exclusive lock is released (0).

P3.10

[Practice | MCQ | 1 Mark]

The lock point of a transaction in 2PL is defined as:

- (A) The point in time when the transaction acquires its very last lock.
- (B) The point when the transaction releases its first lock.
- (C) The time when the transaction commits.
- (D) The point when the first read operation is executed.

Answer: (A) **Solution:** The lock point is the exact moment when the transaction has acquired all the locks it will ever need (the transition point from growing phase to shrinking phase).

4 Topic 4: Logical Reasoning – Series, Coding-Decoding

4.1 10 Previous Year Questions (PYQs)

Q4.1 [GATE GA 2024 (DA/CS) | MCQ | 1 Mark]

Find the missing number in the series: 2, 6, 12, 20, 30, _____, 56.

- (A) 40
- (B) 42
- (C) 44
- (D) 48

Answer: (B) **Solution:** Pattern 1: consecutive products $n(n+1)$: $1 \times 2 = 2$, $2 \times 3 = 6$, $3 \times 4 = 12$, $4 \times 5 = 20$, $5 \times 6 = 30$, $6 \times 7 = 42$, $7 \times 8 = 56$.

Pattern 2: differences are +4, +6, +8, +10, +12, +14. Next is $30 + 12 = 42$.

Q4.2 [GATE GA 2023 | MCQ | 1 Mark]

In a certain code language, if COMPUTER is coded as RFUVQNPC, how is MEDICINE coded in that language?

- (A) EOJDJEFM
- (B) EOJDEJFM
- (C) MFEJDJOE
- (D) EOJDJFEM

Answer: (A) **Solution:** Reverse the word: COMPUTER reversed is RETUPMOC.

Leave the first and last letters unchanged: first letter is R, last letter is C.

Shift intermediate letters by +1: $E \rightarrow F$, $T \rightarrow U$, $U \rightarrow V$, $P \rightarrow Q$, $M \rightarrow N$, $O \rightarrow P$. This gives RFUVQNPC.

For MEDICINE: reversed is ENICIDEM.

First letter E, last letter M. Intermediate letters shifted by +1: $N \rightarrow O$, $I \rightarrow J$, $C \rightarrow D$, $I \rightarrow J$, $D \rightarrow E$, $E \rightarrow F$.

The resulting code is EOJDJEFM.

Q4.3 [GATE GA 2022 | NAT | 1 Mark]

What is the next term in the series: 3, 7, 15, 31, 63, _____? **Answer: 127** **Solution:** Each term is generated by $a_n = 2a_{n-1} + 1$ (or $2^{n+1} - 1$):

$$3 \times 2 + 1 = 7$$

$$7 \times 2 + 1 = 15$$

$$15 \times 2 + 1 = 31$$

$$31 \times 2 + 1 = 63$$

$$63 \times 2 + 1 = 127.$$

Q4.4 [GATE GA 2021 | MCQ | 1 Mark]

If in a code, ROSE is written as 6821, CHAIR is written as 73456, and PREACH is written as 961473, what is the code for SEARCH?

- (A) 214673
- (B) 214763
- (C) 124673
- (D) 216473

Answer: (A) **Solution:** Direct letter-to-digit mapping:

S = 2, E = 1, A = 4, R = 6, C = 7, H = 3.

Thus, SEARCH is coded as 214673.

Q4.5

[GATE GA 2024 | NAT | 2 Marks]

Find the missing number in the sequence: 1, 8, 27, 64, 125, _____. **Answer: 216** **Solution:**

The series consists of cubes of consecutive positive integers n^3 :

$1^3 = 1, 2^3 = 8, 3^3 = 27, 4^3 = 64, 5^3 = 125, 6^3 = 216$.

Q4.6

[GATE GA 2020 | MCQ | 1 Mark]

In a code, if DELHI is coded as CCIDD, how is BOMBAY coded?

(A) AMJXVS

(B) AMJXVT

(C) AMLJVS

(D) AMJXWU

Answer: (A) **Solution:** Examine the letter shifts:

D - 1 = C

E - 2 = C

L - 3 = I

H - 4 = D

I - 5 = D

Pattern: subtract 1, 2, 3, 4, 5, 6 sequentially from the letters of BOMBAY:

B - 1 = A

O - 2 = M

M - 3 = J

B - 4 = X ($28 - 4 = 24$)

A - 5 = V ($27 - 5 = 22$)

Y - 6 = S ($25 - 6 = 19$).

The encoded word is AMJXVS.

Q4.7

[GATE GA 2025 | NAT | 2 Marks]

Find the next number in the series: 2, 3, 5, 9, 17, 33, _____. **Answer: 65** **Solution:** Differences between consecutive terms:

$3 - 2 = 1 = 2^0$

$5 - 3 = 2 = 2^1$

$9 - 5 = 4 = 2^2$

$17 - 9 = 8 = 2^3$

$33 - 17 = 16 = 2^4$.

The next difference is $2^5 = 32$. The next term is $33 + 32 = 65$.

Q4.8

[GATE GA 2023 | MCQ | 1 Mark]

If A = 1, AND = 19, then BAT is equal to:

(A) 23

(B) 21

(C) 22

(D) 20

Answer: (A) **Solution:** Sum of alphabetical letter positions:

A = 1.

$$\text{AND} = 1 + 14 + 4 = 19.$$

$$\text{BAT} = 2 + 1 + 20 = 23.$$

Q4.9

[GATE GA 2026 | NAT | 1 Mark]

What is the 10th term in the arithmetic progression 5, 9, 13, 17, ...? **Answer: 41** **Solution:** First term $a = 5$, common difference $d = 9 - 5 = 4$. The n -th term is $a_n = a + (n - 1)d$. For $n = 10$: $a_{10} = 5 + (10 - 1) \times 4 = 5 + 36 = 41$.

Q4.10

[GATE GA 2019 | MCQ | 1 Mark]

Complete the letter series: AZ, BY, CX, DW, _____.

(A) EV

(B) FU

(C) EU

(D) VE

Answer: (A) **Solution:** First letter advances forward: A, B, C, D $\rightarrow$ E. Second letter moves backward from the end: Z, Y, X, W $\rightarrow$ V. The next term is EV.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | NAT | 2 Marks]

Find the missing number in the series: 4, 9, 25, 49, 121, _____. **Answer: 169** **Solution:** The series consists of squares of consecutive prime numbers:

$$2^2 = 4, 3^2 = 9, 5^2 = 25, 7^2 = 49, 11^2 = 121.$$

The next prime is 13, and its square is $13^2 = 169$.

P4.2

[Practice | MCQ | 1 Mark]

In a certain code, SYSTEM is coded as SYSMET and NEARER is coded as AENRER. How is FRACTION coded in that language?

(A) CARFNOIT

(B) ARFCNOIT

(C) CARFTINO

(D) FRACNOIT

Answer: (A) **Solution:** The word is split into two equal halves of 4 letters each, and each half is reversed:

First half: FRAC reversed becomes CARF.

Second half: TION reversed becomes NOIT.

Combining them yields CARFNOIT.

P4.3

[Practice | NAT | 2 Marks]

What is the missing term in the sequence: 1, 4, 13, 40, 121, _____? **Answer: 364** **Solution:**

Recurrence relation: $a_n = 3a_{n-1} + 1$:

$$1 \times 3 + 1 = 4$$

$$4 \times 3 + 1 = 13$$

$$13 \times 3 + 1 = 40$$

$$40 \times 3 + 1 = 121$$

$$121 \times 3 + 1 = 364.$$

P4.4

[Practice | MCQ | 1 Mark]

If GREAT is coded as 79512 and MONK is coded as 4652, using the single-digit digital root of each letter position ($A = 1, \dots, Z = 26$), how is PEACE coded?

- (A) 75135
- (B) 75134
- (C) 85135
- (D) 76135

Answer: (A) **Solution:** Compute digital root (sum of digits until single digit):

$$P = 16 \rightarrow 1 + 6 = 7.$$

$$E = 5 \rightarrow 5.$$

$$A = 1 \rightarrow 1.$$

$$C = 3 \rightarrow 3.$$

$$E = 5 \rightarrow 5.$$

The resulting code is 75135.

P4.5

[Practice | NAT | 1 Mark]

What is the next number in the alternating series: 2, 12, 4, 10, 6, 8, 8, _____? **Answer: 6** **So-**

lution: The sequence alternates between two interleaved series:

Odd positions: 2, 4, 6, 8, ... (increasing by +2).

Even positions: 12, 10, 8, ... (decreasing by -2).

The next term is at position 8 (an even position): $8 - 2 = 6$.

P4.6

[Practice | MCQ | 2 Marks]

If LIGHT is coded as OJJEW, how is SPARK coded?

- (A) VSDUN
- (B) VRCTN
- (C) VSCUO
- (D) USDUN

Answer: (A) **Solution:** Determine letter shifts:

$$L(+3) = O$$

$$I(+1) = J$$

$$G(+3) = J$$

$$H(-3)? \text{ Let's check: } L(12) \rightarrow O(15) (+3); I(9) \rightarrow J(10) (+1); G(7) \rightarrow J(10) (+3); H(8) \rightarrow E(5) (-3);$$

$$T(20) \rightarrow W(23) (+3).$$

If pattern is +3 on all consonants and +1 on vowels:

$$S(+3) = V$$

$$P(+3) = S$$

$$A(\text{vowel, } +3 \text{ or } +1?): A + 3 = D.$$

$$R(+3) = U$$

$$K(+3) = N.$$

Matches VSDUN where all letters shift by +3.

P4.7

[Practice | NAT | 2 Marks]

Find the sum of the first 8 terms of the Fibonacci sequence 1, 1, 2, 3, 5, 8, ... **Answer: 54**

Solution: The first 8 Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, 21.

$$\text{Sum} = 1 + 1 + 2 + 3 + 5 + 8 + 13 + 21 = 54.$$

$$(\text{Using the identity: } \sum_{i=1}^n F_i = F_{n+2} - 1 = F_{10} - 1 = 55 - 1 = 54.)$$

P4.8

[Practice | MCQ | 1 Mark]

Which term replaces the question mark in the letter series: B4D, E9G, H16J, ??

- (A) K25M
- (B) K25L
- (C) J25M
- (D) L25M

Answer: (A) Solution: First letter: $B(2) \xrightarrow{+3} E(5) \xrightarrow{+3} H(8) \xrightarrow{+3} K(11)$.
 Number: $2^2 = 4, 3^2 = 9, 4^2 = 16, 5^2 = 25$.
 Third letter: $D(4) \xrightarrow{+3} G(7) \xrightarrow{+3} J(10) \xrightarrow{+3} M(13)$.
 The resulting term is K25M.

P4.9

[Practice | NAT | 2 Marks]

Find the missing number in the series: 0, 6, 24, 60, 120, _____. **Answer: 210** **Solution:** The terms follow the formula $n^3 - n$:

$$1^3 - 1 = 0$$

$$2^3 - 2 = 6$$

$$3^3 - 3 = 24$$

$$4^3 - 4 = 60$$

$$5^3 - 5 = 120$$

$$6^3 - 6 = 216 - 6 = 210.$$

P4.10

[Practice | MCQ | 1 Mark]

If in a code, RED is coded as 27 and BLUE is coded as 40, what is the code for GREEN?

- (A) 49
- (B) 50
- (C) 48
- (D) 52

Answer: (A) Solution: Letter position sums:
 $R(18) + E(5) + D(4) = 27$.
 $B(2) + L(12) + U(21) + E(5) = 40$.
 For GREEN: $G(7) + R(18) + E(5) + E(5) + N(14) = 49$.